Typed contracts and schemas

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Grounded Learning Service

WORKBOOK · 1 OF 15

Recall and choose your starting point

> Record your starting explanation for Typed contracts and schemas before opening notes. Show any later correction separately.

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Architecture and data flow

Without opening the old lesson, explain one key idea from Architecture and data flow. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Lesson reference for workbook practice

> Use this reference for the original W1–W3 and W5 exercises. The separately named independent case supplies its own changed facts; use those facts for that attempt before feedback.

Typed contracts define values, required fields and failure outcomes. Runtime **input** still needs validation even when program annotations exist.

Accept exactly room_id as a nonempty string and requested_count as a nonnegative integer excluding bool. Reject extra fields, missing fields and wrong types before **process**ing.

WORKBOOK · 2 OF 15

Practise with fading help

> Complete the supported Typed contracts and schemas trace and record the first step you can explain without help.

Use W2: Trace the worked case for Typed contracts and **schema**s. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

> **The step I completed without help, and the hint I needed**

WORKBOOK · 3 OF 15

Individual reasoning check

> Check the proposed Typed contracts and schemas explanation against the supplied facts before discussing it.

Use the worked case for Typed contracts and **schema**s. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

WORKBOOK · 4 OF 15

Independent transfer case

> Write one first attempt at the reserved independent case before feedback. Repeated copies of that case refer to the same recorded attempt.

Independent case: Apply a new case independently

Classify counts 0, -1, "3", False under the stated contract.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

WORKBOOK · 5 OF 15

W1 Explain the mechanism

> Explain the mechanism. Apply the supplied Typed contracts and schemas facts and keep your reasoning.

Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

My reasoning and evidence

WORKBOOK · 6 OF 15

W2 Trace the worked case

> Trace the worked case. Apply the supplied Typed contracts and schemas facts and keep your reasoning.

Input room_id=B, requested_count=true passes a naive isinstance(value, int) check in Python. Show the intermediate steps and give the conclusion with its scope.

My reasoning and evidence

W3 Audit the limit

> Audit the limit. Apply the supplied Typed contracts and schemas facts and keep your reasoning.

A report omits this qualification: **Schema**-valid **input** can still be unauthorised or semantically unsuitable; subsequent checks remain necessary.

Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

> **My reasoning and evidence**

W4 Apply a new case independently

> Attempt before feedback. If this repeats the selected independent case, reuse that first attempt; otherwise keep a separate record. Apply a new case independently. Apply the supplied Typed contracts and schemas facts and keep your reasoning.

Classify counts 0, -1, "3", False under the stated contract.

> **My reasoning and evidence**

WORKBOOK · 9 OF 15

W5 Design a revealing follow-up

> Design a revealing follow-up. Apply the supplied Typed contracts and schemas facts and keep your reasoning.

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

Do not reuse W4 as a new final test after receiving feedback.

My reasoning and evidence

WORKBOOK · 10 OF 15

Project checkpoint and reflection

> Connect Typed contracts and schemas to Grounded Learning Service, identify a limit and choose the next useful test.

Add a labelled typed contracts and **schema**s **evidence** sheet to Grounded Learning Service. Show how you can explain the **mechanism** and terminology of typed contracts and **schema**s.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

My project change, supporting evidence and remaining limit

WORKBOOK · 11 OF 15

Feedback, return check and learning record

> Record the first repaired step and a different case to revisit after a delay.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

WORKBOOK · 12 OF 15

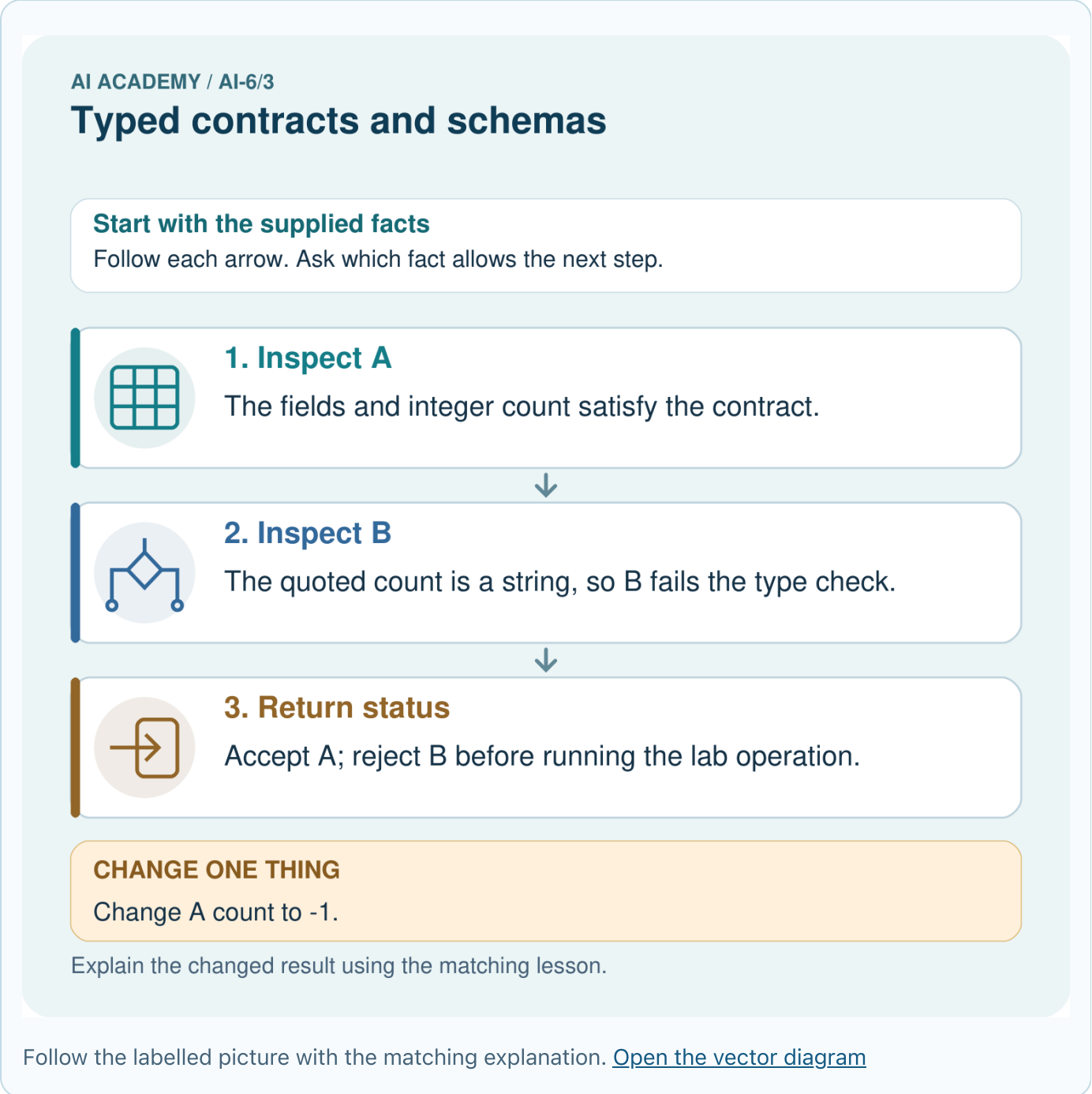
Pictorial case practice: Typed contracts and schemas

> A labelled sample record. A lab tool accepts a string sample_id and an integer count of at

> least zero. Explain the labelled step before checking the changed condition.

A labelled sample record. A lab tool accepts a string sample_id and an integer count of at least zero.

A={sample_id:"jar -8",count:4}; B={sample_id:"jar -9",count:"4"}. No type conversion is allowed.



Follow the labelled picture with the matching explanation. Open the vector diagram

Compare A's count 4 with B's count "4". Which record passes, which gate rejects B, and why does changing A's count to -1 create a different failure?

My picture-based explanation and changed-case prediction

Record the help used

Support used: alone / hint / worked example. This record is practice **evidence**.

Project connection: For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or **evidence** that makes the deciding step valid, so another team can review it.

WORKBOOK · 13 OF 15

Project portfolio milestone

Keep this week's input, deciding step, result and limit together in the Grounded Learning Service record.

Project: Grounded Learning Service

Current milestone: Frame the need. Name the user, the problem, allowed **input**s and a safe simple starting method.

For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or **evidence** that makes the deciding step valid, so another team can review it.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	
One error or limit and the next test	

> **Explain what this week adds to your project. Reuse last week's record rather than starting again.**

WORKBOOK · 14 OF 15

Supported practice and learning evidence

> Try the supported questions before comparing with hints and feedback. Record what you repaired and the evidence you retained.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

A labelled sample record. A lab tool accepts a string sample_id and an integer count of at least zero.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Accept A; reject B before running the lab operation.

1. Inspect A

2. Inspect B

3. Return status

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Return status Accept A; reject B before running the lab operation. A **schema** checks declared structure.

Looking like a number is different from having the required numeric type.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: Change A count to -1. Which conclusion is supported by the supplied facts?

1. This one example guarantees correct results for every future **input**.

2. Reject A on the non-negative value rule.

3. We can decide the result without examining the changed **input** or the procedure.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Reject A on the non-negative

value rule. The type is correct but the value is outside its accepted range.

Repair a missing building block

Review Architecture and data flow (ai-6/2). Goal: Explain the **mechanism** and terminology of architecture and data flow. Short practice: Label what moves along the library request, retrieval and drafting arrows.

Where does 17:30 come from, and why must the service stop if the retrieval arrow is disconnected?. Return to this week and explain what you changed.

Predict, run and debug

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

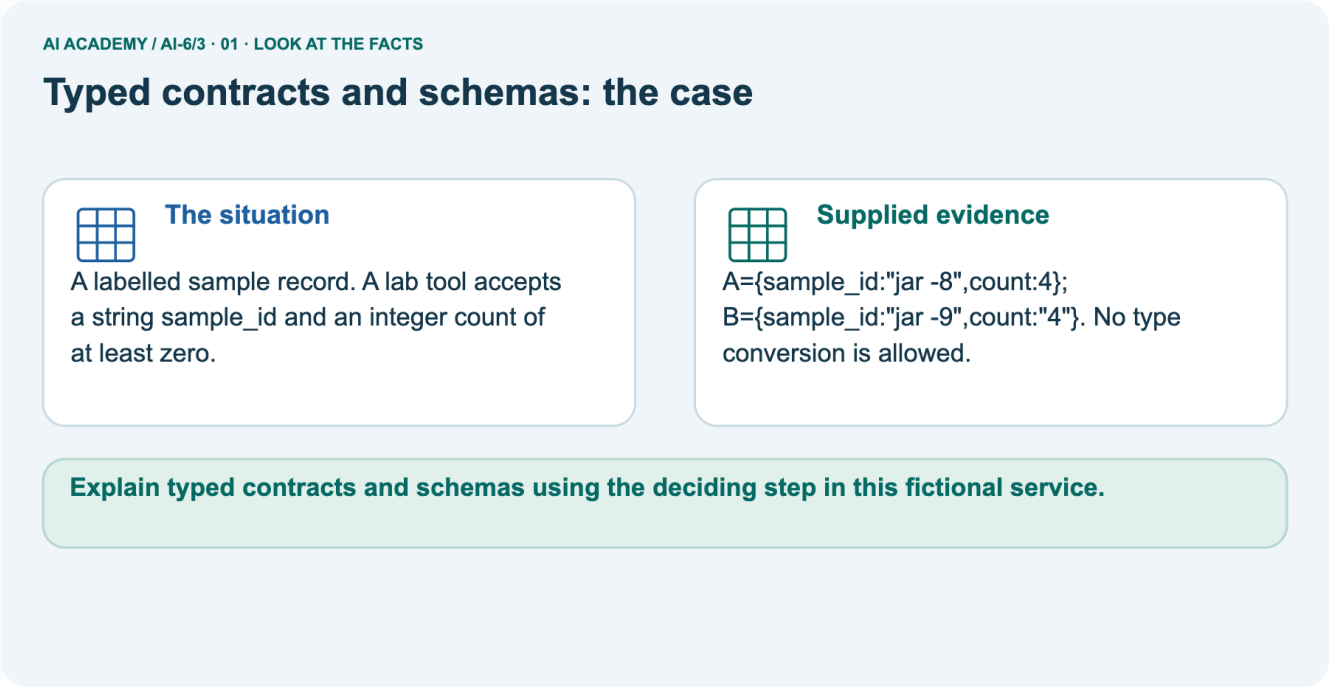
Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test

Visual concept journey

> Explain each Typed contracts and schemas picture from its labels, then record a relevant limit.

01. Typed contracts and schemas: the case



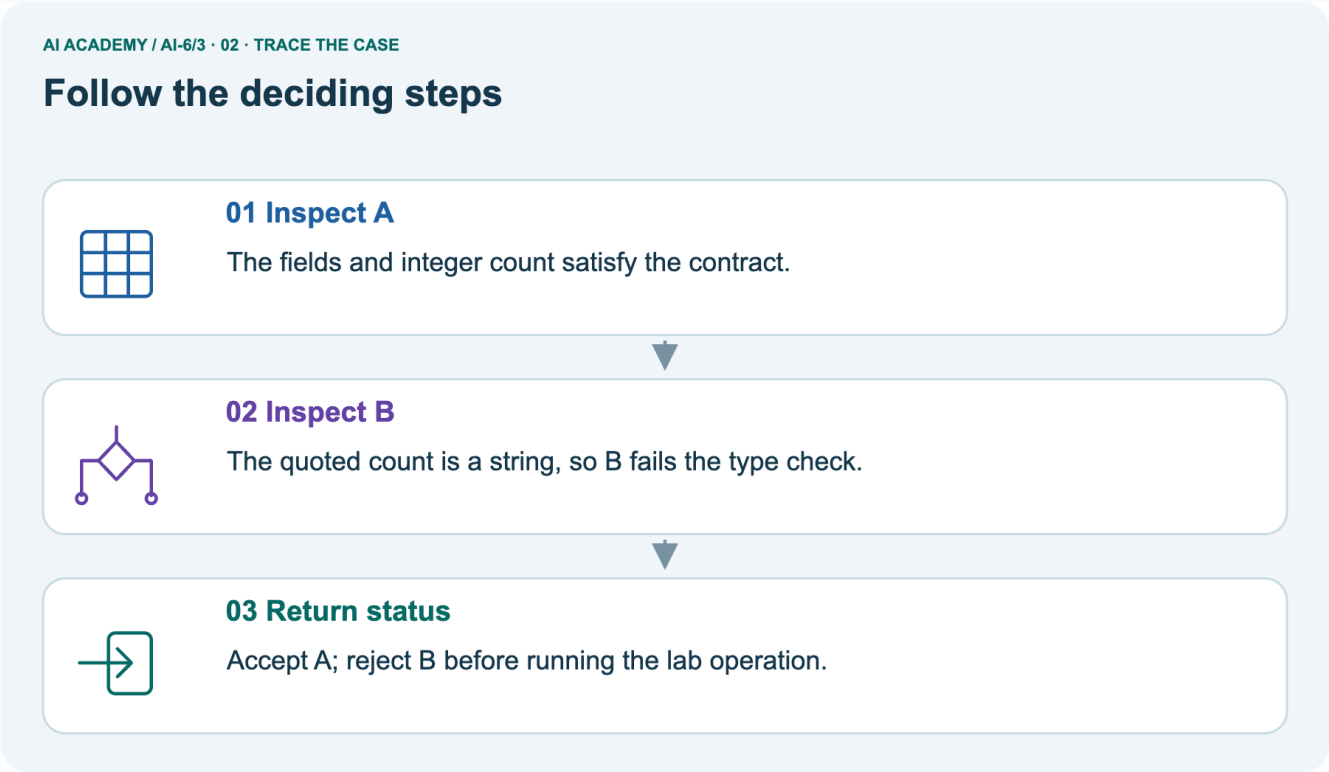
Follow the labelled picture with the matching explanation. Open the vector diagram

A labelled sample record. A lab tool accepts a string sample_id and an integer count of at least zero. Supplied facts: A={sample_id:"jar -8",count:4}; B={sample_id:"jar -9",count:"4"}. No type conversion is allowed.

> Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



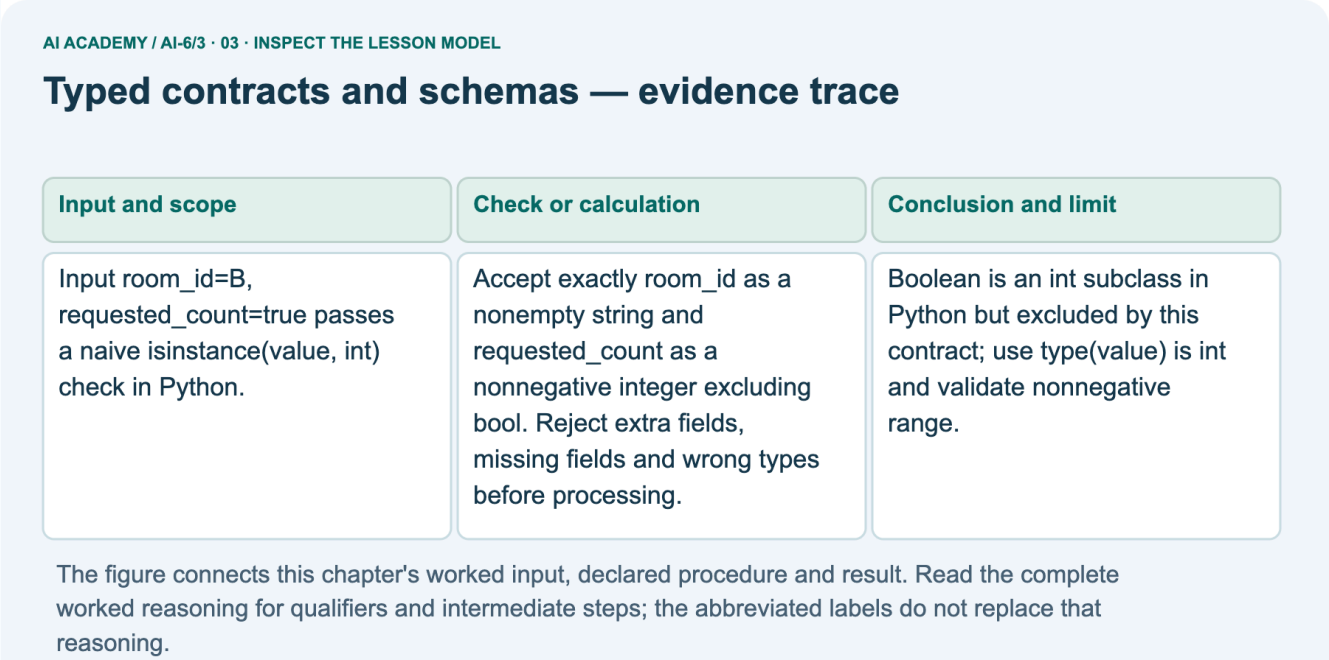
Follow the labelled picture with the matching explanation. Open the vector diagram

Inspect A: The fields and integer count satisfy the contract. → Inspect B: The quoted count is a string, so B fails the type check. → Return status: Accept A; reject B before running the lab operation.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Typed contracts and schemas — evidence trace



Follow the labelled picture with the matching explanation. Open the vector diagram

Typed contracts and **schema**s — **evidence** trace.

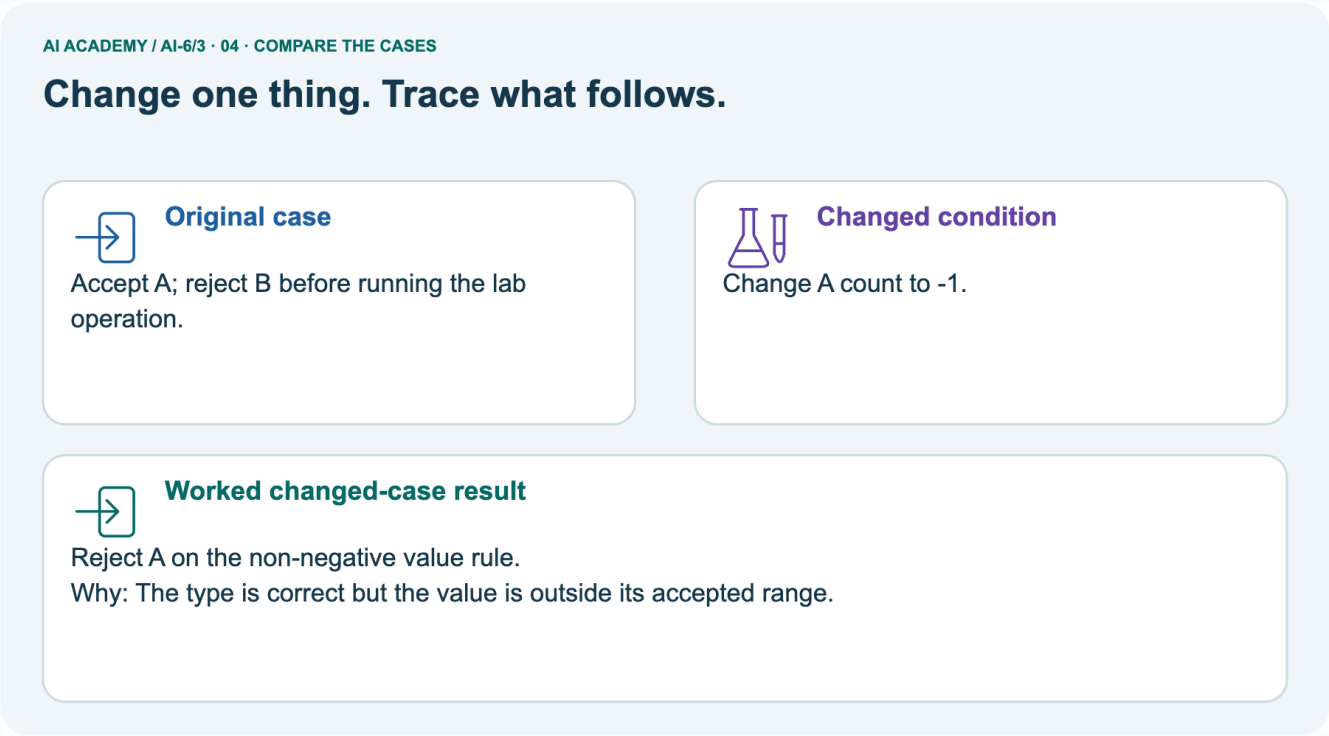
Input room_id=B, requested_count=true passes a naive isinstance(value, int) check in Python. | Accept exactly room_id as a nonempty string and requested_count as a nonnegative integer excluding bool.

Reject extra fields, missing fields and wrong types before **process**ing. | Boolean is an int subclass in Python but excluded by this contract; use type(value) is int and validate nonnegative range.

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

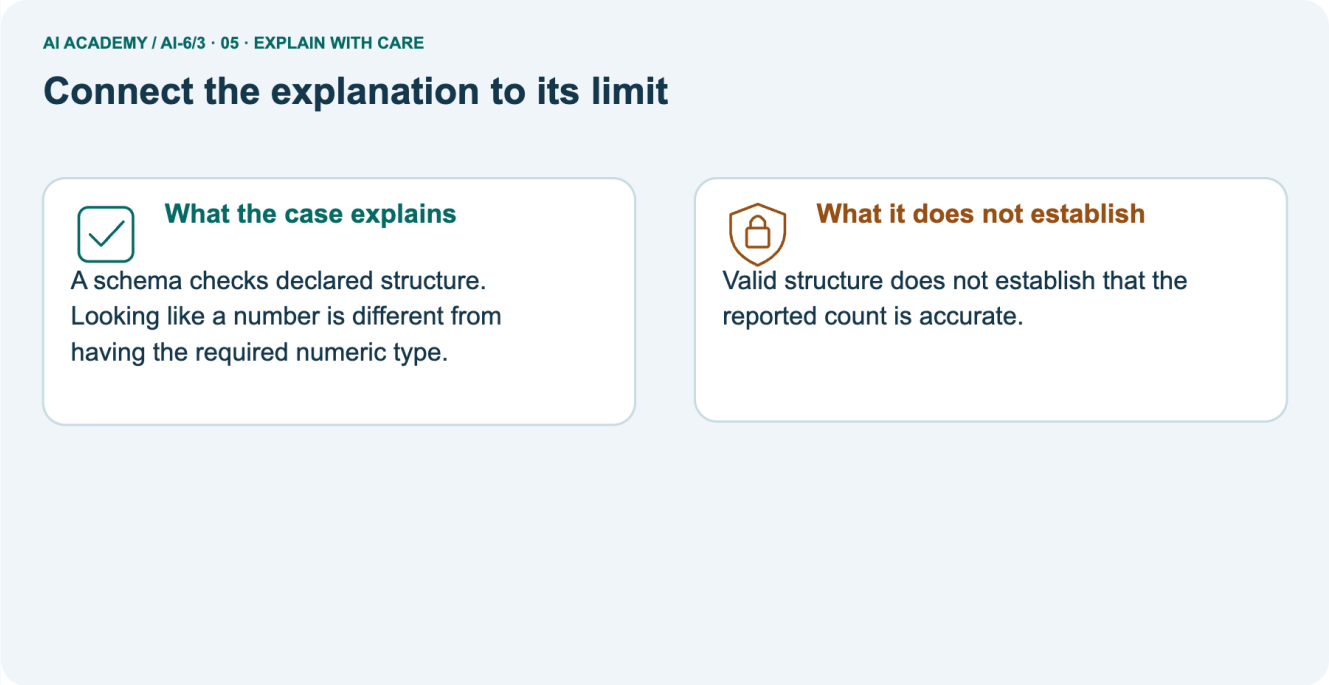
Original case: Accept A; reject B before running the lab operation. Changed condition: Change A count to -1. Expected result: Reject A on the non-negative value rule.

Reason: The type is correct but the value is outside its accepted range.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



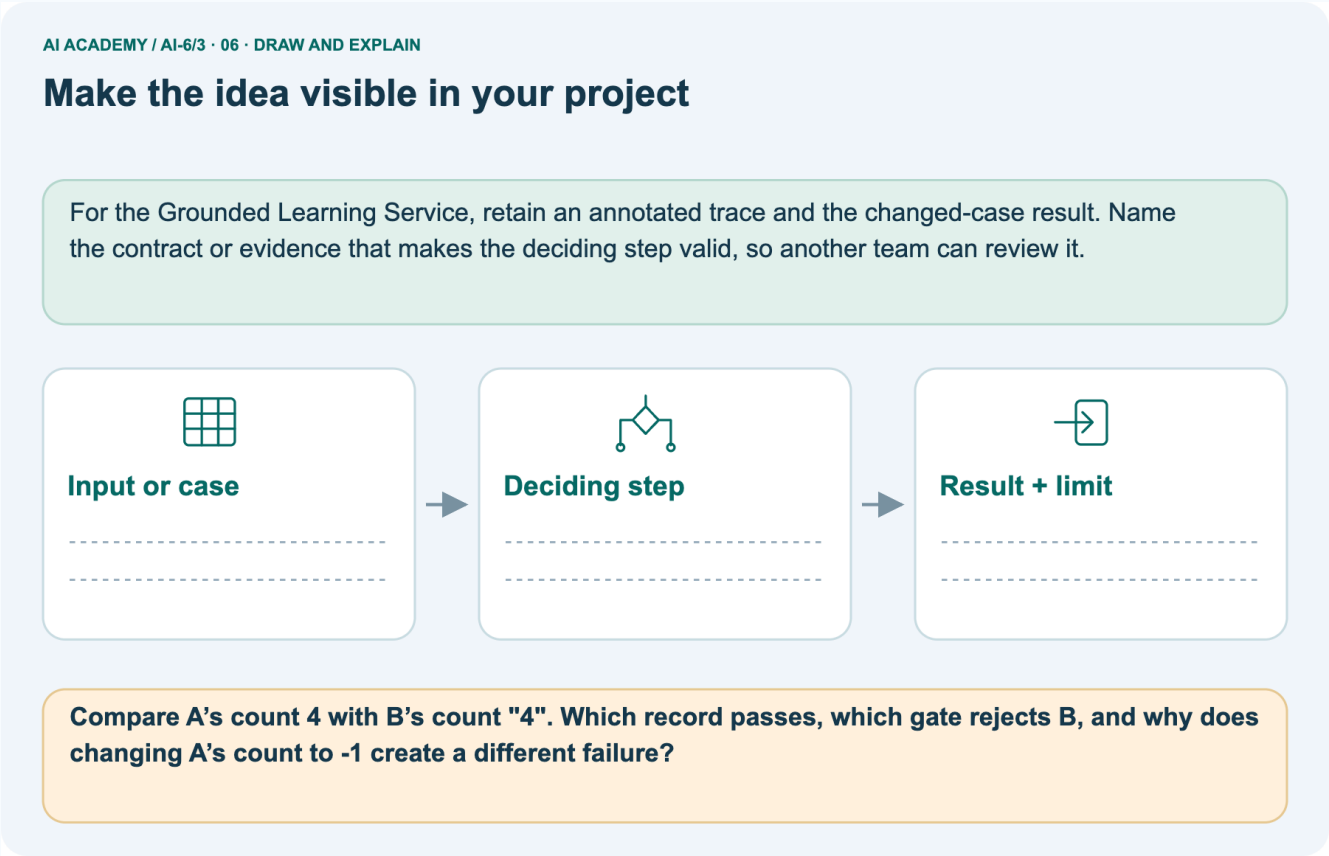
Follow the labelled picture with the matching explanation. Open the vector diagram

A **schema** checks declared structure. Looking like a number is different from having the required numeric type. Limit: Valid structure does not establish that the reported count is accurate.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or **evidence** that makes the deciding step valid, so another team can review it.

Draw your **input**, deciding step, result and limit. Supported question: Compare A's count 4 with B's count "4". Which record passes, which gate rejects B, and why does changing A's count to -1 create a different failure?

Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.